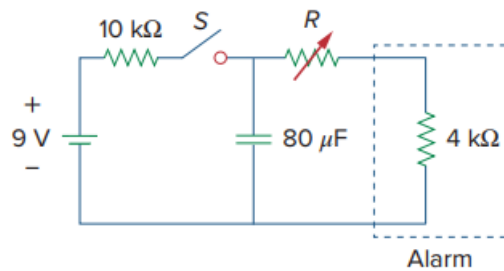


Problem

The RC circuit in Fig. 7.74 is designed to operate an alarm which activates when the current through it exceeds $90\ \mu\text{A}$. If $0 \leq R \leq 6\ \text{k}\Omega$, find the range of the time delay that the variable resistor can create.

Fig. 7.74:



Step-by-step solution

Step 1 of 10

Refer to Figure 7.74 in the textbook.

The capacitor is not charged initially, thus, $v_c(0) = 0$.

For $R = 0$, redraw the circuit as shown in Figure 1.

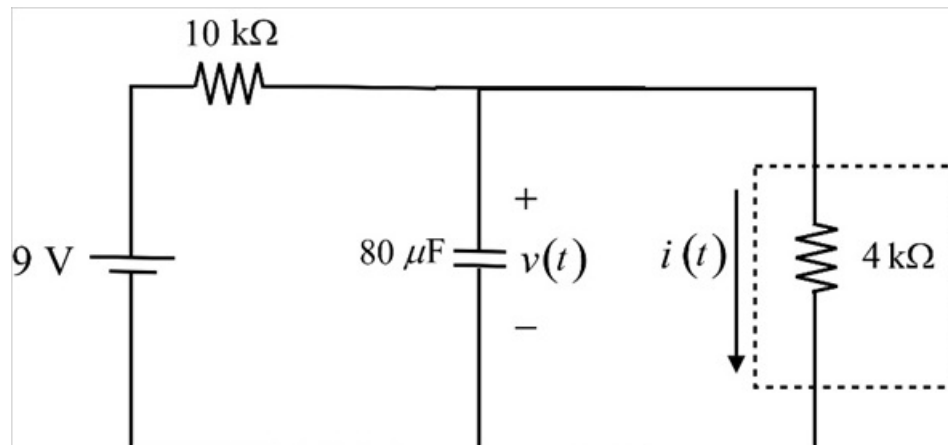


Figure 1

Step 2 of 10

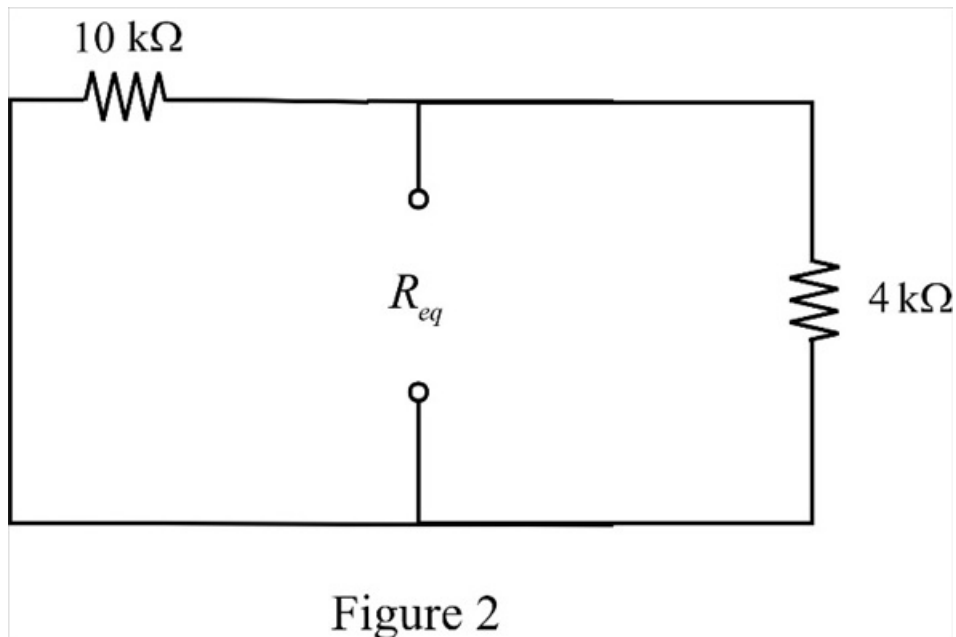
For $t = \infty$, the capacitor becomes open circuited.

Calculate the voltage across the capacitor with voltage divider rule as follows:

$$v(\infty) = (9) \left[\frac{4 \times 10^3}{(4 + 10) \times 10^3} \right]$$

$$= 2.571 \text{ V}$$

To calculate the time constant, remove the capacitor and short circuit the 9 V voltage source, as shown in Figure 2.



Step 3 of 10

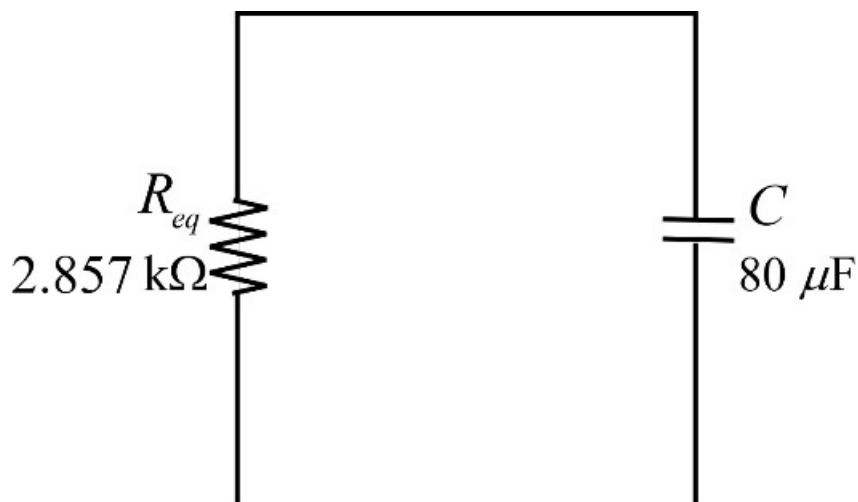
The equivalent resistance is the parallel combination of $10 \text{ k}\Omega$ and $4 \text{ k}\Omega$ resistors.

$$R_{eq} = (10 \times 10^3) \parallel (4 \times 10^3)$$

$$= \frac{(10 \times 10^3)(4 \times 10^3)}{(10 \times 10^3) + (4 \times 10^3)}$$

$$= 2.857 \text{ k}\Omega$$

Connect the capacitor (C) in series with the equivalent resistance (R_{eq}), as shown in Figure 3.



Step 4 of 10

The time constant is the product of the equivalent resistance and the capacitance.

$$\tau = R_{eq} C$$

Substitute $2.857 \text{ k}\Omega$ for R_{eq} and $80 \text{ }\mu\text{F}$ for C .

$$\begin{aligned}\tau &= (2.857 \times 10^3)(80 \times 10^{-6}) \\ &= 0.2286 \text{ s}\end{aligned}$$

The voltage at time t is represented as follows:

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t_o}{\tau}}$$

Substitute 2.571 V for $v(\infty)$, 0 V for $v(0)$, 0.2286 s for τ .

$$\begin{aligned}v(t) &= 2.571 + [0 - 2.571]e^{-\frac{t_o}{0.2286}} \\ &= 2.571 \left(1 - e^{-\frac{t_o}{0.2286}}\right) \text{ V}\end{aligned}$$

The current through the $4 \text{ k}\Omega$ resistor or the alarm is as follows:

$$i(t) = \frac{v(t)}{4 \times 10^3}$$

Substitute $2.571 \left(1 - e^{-\frac{t_o}{0.2286}}\right)$ for $v(t)$.

$$\begin{aligned}i(t) &= \frac{2.571 \left(1 - e^{-\frac{t_o}{0.2286}}\right)}{4 \times 10^3} \\ &= 0.643 \left(1 - e^{-\frac{t_o}{0.2286}}\right) \text{ mA}\end{aligned}$$

Step 5 of 10

The minimum allowed current through the alarm is $90 \text{ }\mu\text{A}$; substitute $90 \text{ }\mu\text{A}$ for $i(t)$.

$$90 \times 10^{-6} = 0.643 \left(1 - e^{-\frac{t_o}{0.2286}}\right) \times 10^{-3}$$

$$e^{-\frac{t_o}{0.2286}} = 1 - \left(\frac{90}{0.643} \times 10^{-3}\right)$$

$$t_o = 34.47 \text{ ms}$$

Step 6 of 10

For $R = 6\text{ k}\Omega$, redraw the circuit as shown in Figure 4.

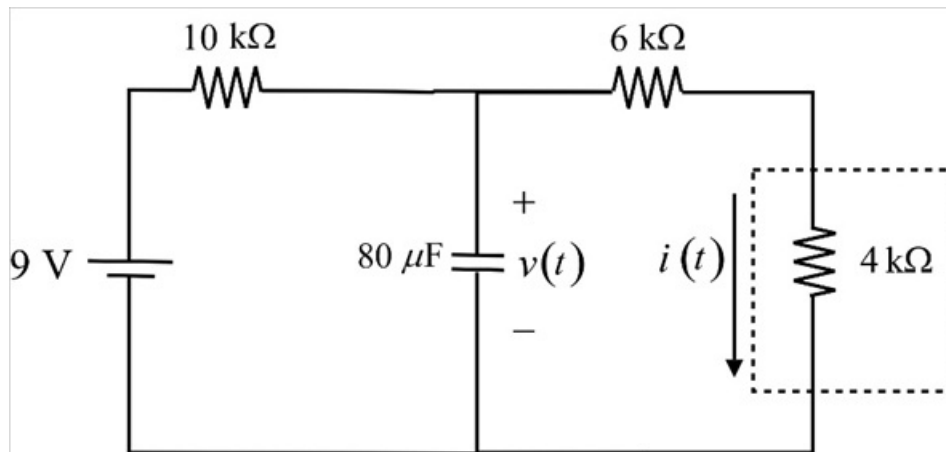


Figure 4

Step 7 of 10

For $t = \infty$, the capacitor becomes open circuited.

Calculate the voltage across the capacitor with voltage divider rule as follows:

$$v(\infty) = (9) \left[\frac{(6+4) \times 10^3}{(10+6+4) \times 10^3} \right]$$
$$= 4.5\text{ V}$$

To calculate the time constant, remove the capacitor and short circuit the 9 V voltage source, as shown in Figure 5.

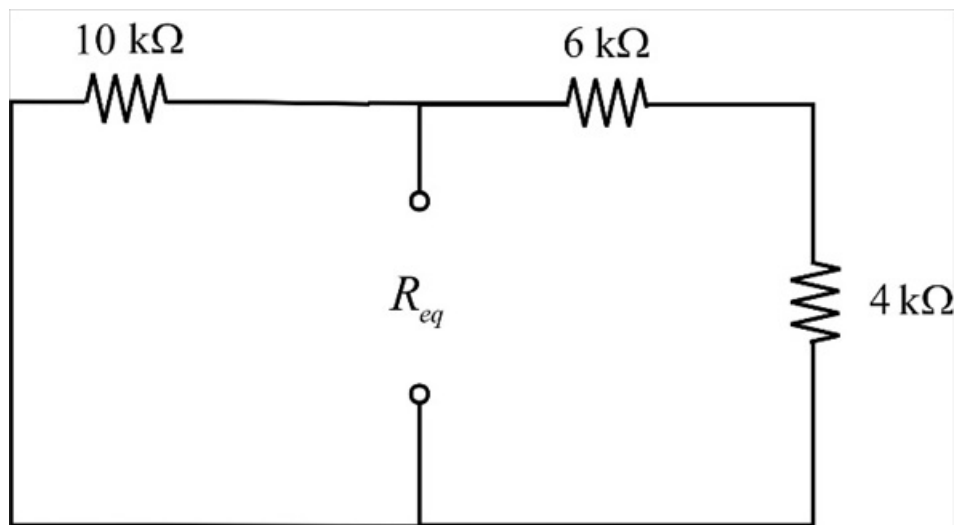


Figure 5

Step 8 of 10

The equivalent resistance is the parallel combination of $10\text{ k}\Omega$ resistor with the series combination of $6\text{ k}\Omega$ and $4\text{ k}\Omega$.

$$\begin{aligned} R_{eq} &= (10 \times 10^3) \parallel ((6 + 4) \times 10^3) \\ &= \frac{(10 \times 10^3)(10 \times 10^3)}{(10 \times 10^3) + (10 \times 10^3)} \\ &= 5\text{ k}\Omega \end{aligned}$$

Connect the capacitor (C) in series with the equivalent resistance (R_{eq}), as shown in Figure 6.

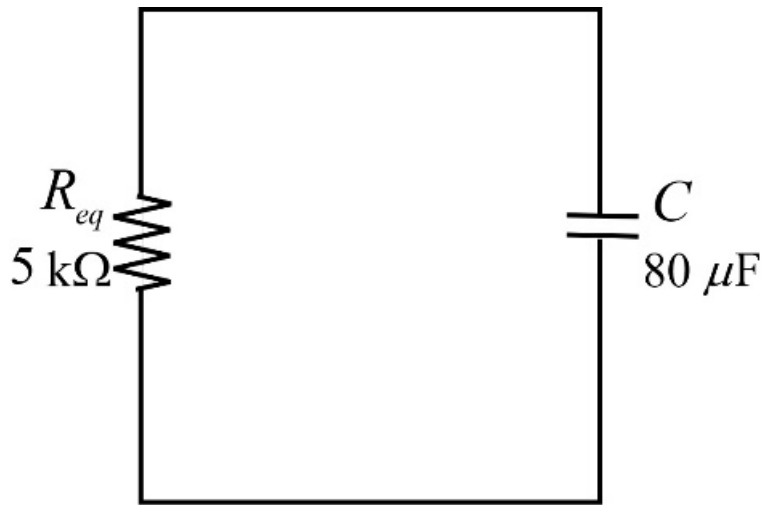


Figure 6

Step 9 of 10

The time constant is the product of the equivalent resistance and the capacitance.

$$\tau = R_{eq} C$$

Substitute $5\text{ k}\Omega$ for R_{eq} and $80\text{ }\mu\text{F}$ for C .

$$\begin{aligned} \tau &= (5 \times 10^3)(80 \times 10^{-6}) \\ &= 0.4\text{ s} \end{aligned}$$

The voltage at time t is represented as follows:

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t_0}{\tau}}$$

Substitute 4.5 V for $v(\infty)$, 0 V for $v(0)$, 0.4 s for τ .

$$\begin{aligned} v(t) &= 4.5 + [0 - 4.5]e^{-\frac{t_0}{0.4}} \\ &= 4.5 \left(1 - e^{-\frac{t_0}{0.4}} \right) \text{ V} \end{aligned}$$

Step 10 of 10

Calculate the current through the combination $4\text{ k}\Omega$ and $6\text{ k}\Omega$ resistors or the alarm as follows:

$$i(t) = \frac{v(t)}{(6+4) \times 10^3}$$

$$= \frac{v(t)}{10 \times 10^3}$$

Substitute $4.5\left(1 - e^{-\frac{t_o}{0.4}}\right)$ for $v(t)$.

$$i(t) = \frac{4.5\left(1 - e^{-\frac{t_o}{0.4}}\right)}{10 \times 10^3}$$

$$= 0.45\left(1 - e^{-\frac{t_o}{0.4}}\right) \text{ mA}$$

The minimum allowed current through the alarm is $90\text{ }\mu\text{A}$; substitute $90\text{ }\mu\text{A}$ for $i(t)$.

$$90 \times 10^{-6} = 0.45\left(1 - e^{-\frac{t_o}{0.4}}\right) \times 10^{-3}$$

$$e^{-\frac{t_o}{0.4}} = 1 - \left(\frac{90}{0.45} \times 10^{-3}\right)$$

$$t_o = 89.26 \text{ ms}$$

Hence, the range of time delay is from 34.47 ms to 89.26 ms .